

1 Highlights

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- 6 • Land-derived Landsat/Sentinel-2 harmonization is invalid over dark
7 oligotrophic water
- 8 • Sensors stay 99.8% separable after harmonization; intercepts exceed
9 signal 70-fold
- 10 • Campaign, not station, is the blocking unit in campaign-based moni-
11 toring programmes
- 12 • Honest Secchi retrieval: $R^2=0.574$, RMSE 1.86 m over 812 real match-
13 ups
- 14 • Conformal intervals are calibrated marginally but under-cover in the
15 clearest water

Harmonization and validation limits of satellite Secchi retrieval in Lake Titicaca

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Abstract

Monitoring water transparency in high-altitude oligotrophic lakes is critical for freshwater management, yet it remains constrained by the sparsity of field data. While multi-mission satellite archives (e.g., harmonized Sentinel-2 and Landsat 8/9) and machine learning offer a synoptic solution, current practices often import assumptions from turbid-water literature that fail in clear-water, campaign-based settings. We assemble 1002 satellite–field match-ups against twelve years of limnological monitoring in Lake Titicaca to address these failures and define a defensible retrieval protocol. First, we demonstrate that standard land-derived harmonization coefficients (e.g., Roy et al., 2016) are invalid over dark water, as their intercepts exceed the native Landsat signal by factors of 1.4 to 69.8, leaving sensors 99.8% separable; we propose a classifier two-sample test to catch this failure. Second, we show that standard station-blocked cross-validation fails to prevent spatial leakage in campaign-based sampling: under campaign-blocked folds, a station-mean null model still explains $R^2 = 0.208$ of the variance. We introduce a null-model diagnostic to identify the correct blocking unit (the campaign). Finally, under this strict design, we train a Random Forest regressor to retrieve Secchi disk

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depth ($R^2 = 0.574$, RMSE 1.86 m) and attach split-conformal prediction intervals. We reveal that while marginally calibrated (89.4% coverage at 90% nominal), conformal intervals structurally under-cover in the clearest water (71.3%). Our findings provide a rigorous blueprint for evaluating machine-learning water quality products in large, oligotrophic lakes.

Keywords: Lake Titicaca, Secchi disk depth, Sentinel-2, Landsat, cross-sensor harmonization, conformal prediction

1. Introduction

Large lakes are sentinels of environmental change, but continuous monitoring of remote, high-altitude systems is constrained by sparse field infrastructure (Williamson et al., 2009). Lake Titicaca (3,810 m a.s.l.), the largest tropical lake in the world by volume, is a critical transboundary freshwater reserve whose water quality is documented only by infrequent field campaigns (OEFA, 2016; Signayro Mamani et al., 2022), even as eutrophication advances in its near-shore bays (Heredia et al., 2022; Lanza et al., 2024). Satellite remote sensing offers a synoptic solution (Gholizadeh et al., 2016; Topp et al., 2020). Specifically, water transparency—quantified by the Secchi disk depth (Z_{SD})—is an integrative indicator of water clarity strongly linked to light attenuation (Lee et al., 2015). Because Sentinel-2 MSI and Landsat 8/9 OLI provide complementary revisit times, their joint use as a virtual constellation coupled with machine learning is now the standard practice for mapping water quality (Pahlevan et al., 2020; Cao et al., 2022).

Although these virtual constellations work well in turbid, productive “Case-2” waters, they fail to transfer reliably to clear, oligotrophic high-

altitude lakes due to two unresolved methodological assumptions. First, the cross-sensor harmonization coefficients required to merge Landsat and Sentinel-2 (e.g., Roy et al., 2016) were fitted on terrestrial targets. Because terrestrial surfaces are orders of magnitude brighter than clear water, their additive intercepts overwhelm the over-water signal. While near-infrared bias over water has been noted (Pahlevan et al., 2019), the magnitude of this mismatch and its effect on machine-learning retrievals remain unquantified. Second, to prevent spatial autocorrelation from inflating apparent accuracy, previous methods routinely block cross-validation by monitoring station (Roberts et al., 2017). However, they assume rather than test that station identity is the primary source of data leakage. In campaign-based monitoring programmes, where an entire lake is sampled within a few days under identical conditions, station-blocking may retain severe leakage and falsely inflate performance metrics.

In this paper, we propose a rigorous validation and retrieval pipeline to overcome these challenges, using Lake Titicaca as our testing ground. Our innovation is replacing assumed methodological defaults with data-driven diagnostics. Specifically, we assemble 1002 real satellite–field match-ups and apply a classifier two-sample test (adversarial validation) (Lopez-Paz and Oquab, 2017) to diagnose harmonization failure. Furthermore, we introduce a null-model diagnostic to identify the true blocking unit for cross-validation. Finally, we train a Random Forest regressor under the corrected design and attach split-conformal prediction intervals (Angelopoulos and Bates, 2023) to quantify per-prediction uncertainty.

In contrast to previous water-quality mapping efforts (Ayala Izurieta

et al., 2023; Maligaya et al., 2024), our study makes three explicit contributions:

1. **Quantification of Harmonization Limits:** We show that land-derived harmonization intercepts exceed the native Landsat signal by up to 69.8-fold over clear water. Consequently, a classifier can identify the source sensor with 99.8% accuracy from harmonized bands, proving that the model learns a sensor label rather than unified optics.
2. **Null-Model Validation Design:** We demonstrate that under campaign-blocked folds, a station-mean null model still achieves $R^2 = 0.208$, proving that station-blocking fails to prevent leakage. We establish the campaign as the required blocking unit for this class of data.
3. **Defensible Retrieval with Conformal Uncertainty:** Under this strict design, our model achieves an honest R^2 of 0.574 (RMSE 1.86 m). We further reveal that while split-conformal intervals are marginally calibrated (89.4% coverage), they structurally under-cover extreme values in the clearest water, highlighting a critical limitation for operational monitoring.

2. Materials and methods

2.1. Study area

Lake Titicaca comprises three limnologically distinct zones (Fig. 1): the large, deep, oligotrophic Lago Mayor; the shallower Lago Menor (Wiñaymarca) to the south-east; and Bahía de Puno, which receives municipal and agricultural inputs from the city of Puno and is the most eutrophic sector.

110 Measured transparency spans 1.5 m in near-shore turbid water to 16.5 m in
111 the open oligotrophic basin.

112 2.2. *In-situ data*

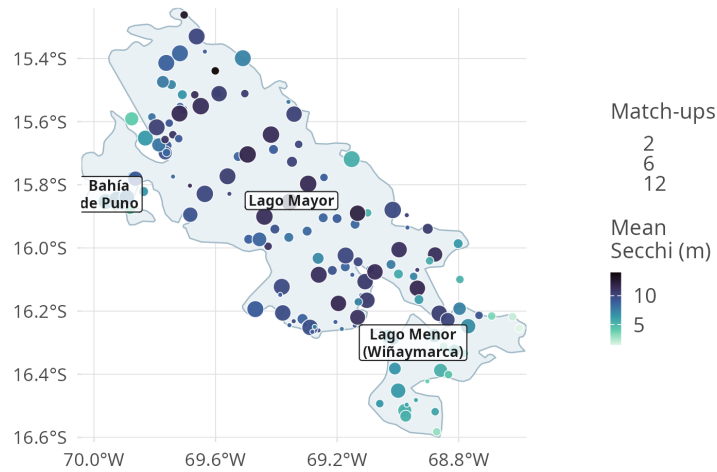
113 We used 881 field measurements of Secchi transparency, with co-measured
114 chlorophyll-*a*, suspended solids, temperature, pH, dissolved oxygen and nu-
115 trients, from the limnological monitoring campaigns of IMARPE (Instituto
116 del Mar del Perú) on Lake Titicaca, spanning 2011–2024 across 160 sta-
117 tion identifiers distributed over the whole lake. Six of those identifiers are
118 a 2017 naming variant of a station already recorded under another label,
119 so the records describe 154 distinct stations; we retain the identifiers as
120 issued, which makes station-blocked validation marginally more permissive
121 and therefore does not favour the argument of Section 2.6. Of the 881 mea-
122 surements, 734 include a valid Secchi reading. The July 2019 campaign (71
123 stations) is published by Siguayro Mamani and Franco (2022); the remaining
124 years derive from the same IMARPE programme. As a provenance check, our
125 2019 zone-level means reproduce the published values almost exactly (Lago
126 Mayor conductivity 1513.4 vs. 1513.5 $\mu\text{S cm}^{-1}$; Lago Menor chlorophyll-*a*
127 1.52 vs. 1.52 mg m^{-3}).

128 Three properties of this record constrain everything that follows, and we
129 state them here rather than in the limitations (Fig. A.5).

130 First, the monitoring programme is itself almost entirely a dry-season
131 programme. The 2011–2024 record contains 13 campaigns with Secchi read-
132 ings, of which exactly one — December 2012, 72 measurements — falls in the
133 austral wet season (December–March). That campaign predates both sen-
134 sors used here: Landsat 8 became operational in May 2013 and Sentinel-2A

(a) Limnological monitoring network (IMARPE)

IMARPE/ALT stations with satellite-field match-ups (2013–2024), coloured by mean measured Secchi depth. The trophic gradient is visible in the field data alone



(b) Sampling effort is very uneven

Each point is a station. 143 stations, but most were visited in very few campaigns, limiting the power of any station-based analysis

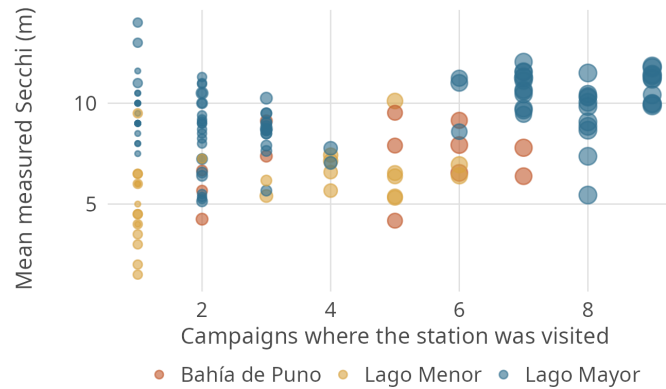


Figure 1: Study area and monitoring network. (a) IMARPE/ALT stations with a satellite–field match-up, coloured by mean measured Secchi depth over the Natural Earth lake boundary; the trophic gradient is evident in the field data alone. (b) Sampling effort per station: most stations were visited in few campaigns, which limits the power of any station-level analysis.

135 in June 2015. Consequently every match-up in our dataset falls between July
136 and October, and this is a property of the field programme rather than of
137 our processing: no choice of match-up window or cloud threshold can recover
138 wet-season pairs that have no contemporaneous imagery. We return to what
139 this means for operational use in Section 4.3.

140 Second, sampling is episodic: campaigns exist for 2013–2024 with no
141 campaigns in 2020, 2021, 2023.

142 Third, the reference measurement is coarse: 73 distinct Secchi values
143 occur in 812 match-ups, because disk readings are recorded to the nearest
144 half metre.

145 No synthetic or spectrally derived labels were used at any stage.

146 *2.3. Satellite data and cross-sensor harmonization*

147 Sentinel-2 MSI (harmonized surface reflectance) and Landsat 8/9 OLI
148 (Collection 2, Level-2; Vermote et al., 2016) were accessed through Google
149 Earth Engine (Gorelick et al., 2017). Clouds were masked with the Sentinel-
150 2 Scene Classification Layer plus cloud probability, and with the Landsat
151 QA_PIXEL bitmask (dilated cloud, cloud and shadow). The cirrus flag was
152 not applied, as it is near-permanently set at 3810 m and would mask the
153 entire archive. We retained the six bands common to both sensors (blue,
154 green, red, NIR, SWIR1, SWIR2).

155 Landsat surface reflectance was first harmonized to Sentinel-2 equivalent
156 reflectance with the per-band linear coefficients of Roy et al. (2016), of the
157 form $\rho_{\text{MSI}} = a + b \rho_{\text{OLI}}$. Section 3.1 evaluates whether this transform is valid
158 over the water of Lake Titicaca and derives an alternative. For that evalu-
159 ation we identified every in-situ event observed by both sensors within the

160 match-up window — 270 paired station–date observations — and regressed
161 Sentinel-2 on harmonized Landsat reflectance band by band.

162 *2.4. Satellite–field match-ups*

163 For every in-situ measurement we extracted the cloud-masked median
164 reflectance within a 30 m buffer of the station coordinate, from imagery
165 acquired within ± 10 days. The 30 m buffer matches the coarser Landsat pixel
166 and averages residual geolocation error and sensor noise over several pixels
167 while remaining in open water, away from shoreline mixing. The ± 10 -day
168 window balances the scarce cloud-free image availability at 3810 m against
169 the slow temporal change of transparency in this large, deep lake, and is
170 consistent with windows used in operational Z_{SD} retrieval (Kloiber et al.,
171 2002; Zhang et al., 2022).

172 This yielded 1002 match-ups (340 Sentinel-2, 662 Landsat), of which 812
173 have a Secchi value, across 143 stations and 120 campaign dates. Because
174 both sensors can observe the same field event, 272 match-ups are cross-sensor
175 pairs of a single in-situ measurement; collapsing them leaves 540 independent
176 events, and Section 3.2 verifies that accuracy is unchanged, confirming the
177 result is not an artefact of pseudo-replication.

178 *2.5. Features and model*

179 Twelve sensor-agnostic features were computed from the six common
180 bands: the bands themselves plus NDWI, NDTI and the ratios blue/green,
181 green/blue, red/green and blue/red. We trained a Random Forest regres-
182 sor (Breiman, 2001). Reflectance was scaled with a robust (median/IQR)

183 scaler fitted only on the training partition of each fold. All models were
184 implemented with scikit-learn (Pedregosa et al., 2011).

185 Model choice was settled empirically rather than assumed. We com-
186 pared Random Forest, Extremely Randomized Trees, XGBoost (Chen and
187 Guestrin, 2016) and LightGBM, each with hyperparameters optimized by
188 Optuna (Akiba et al., 2019) inside *nested* cross-validation — tuning on an
189 inner campaign-blocked split of each outer training partition, so no outer fold
190 ever informs its own hyperparameters — and benchmarked the resulting gain
191 against the spread across random seeds. Section 3.4 reports the outcome.

192 2.6. Validation design and null models

193 Rather than a single split, we evaluate a hierarchy of increasingly de-
194 manding designs (Roberts et al., 2017): random K -fold; GroupKFold blocked
195 by station; GroupKFold blocked by *campaign date*; GroupKFold blocked by
196 year; a temporal hold-out; and leave-one-zone-out, in which an entire trophic
197 zone is withheld.

198 Crucially, we do not assume which blocking unit matters. We benchmark
199 each design against null models that predict, respectively, the training-set
200 mean of the zone, of the station and of the campaign date. A blocking unit
201 is doing real work if, and only if, the corresponding null model collapses
202 under it. This diagnostic, rather than convention, determines our primary
203 design.

204 Performance is reported as out-of-fold R^2 , RMSE and MAE, and bench-
205 marked against a classical log-linear blue/green band-ratio algorithm in the
206 standard form $\ln Z_{SD} \sim \ln(\rho_{\text{blue}}/\rho_{\text{green}})$ (Kloiber et al., 2002), with predic-
207 tions bounded to the physically observable range, and against a multiband

208 linear model.

209 2.7. Uncertainty and interpretation

210 We attach a per-prediction interval using split-conformal prediction (An-
211 gelopoulos and Bates, 2023): within each fold, a calibration subset of the
212 training partition yields the $(1 - \alpha)$ quantile of absolute residuals, defining
213 a symmetric interval with finite-sample marginal coverage guarantees (Ro-
214 mano et al., 2019). Because a single nominal level can be met by chance, we
215 report empirical coverage at 50, 80, 90 and 95%, and also coverage condi-
216 tional on measured transparency, since marginal calibration does not imply
217 conditional calibration.

218 The model is interpreted with SHAP (Lundberg and Lee, 2017; Lundberg
219 et al., 2020), computed *out of fold* — each observation explained by the model
220 that did not see it — rather than on a model fitted to the whole dataset, which
221 inflates the apparent importance of memorized features. Trends in annual
222 median in-situ transparency were tested with Mann–Kendall and Sen’s slope
223 (Mann, 1945; Sen, 1968) on the full 2011–2024 in-situ record, not on the
224 match-up subset.

225 3. Results

226 3.1. Land-derived harmonization is invalid over oligotrophic water

227 The Roy et al. (2016) transforms carry additive intercepts of 0.0116–
228 0.0448 in reflectance units. Over Lake Titicaca the native Landsat over-water
229 signal is far smaller than this (Table A.2, Fig. 2a). In the near-infrared the
230 median native reflectance is effectively zero, so the intercept exceeds the

231 signal by a factor of 69.8; in SWIR1 the factor is 23.2, in red 10.6, and even
 232 in blue — the brightest water band — 2.5. The harmonized Landsat bands
 233 are consequently dominated by a constant inherited from a land calibration:
 234 their medians exceed the corresponding Sentinel-2 medians by 1.6-fold in
 235 green and 15.4-fold in the near-infrared.

236 A harmonization that succeeded would make the two sensors statistically
 237 indistinguishable over the same water. They are not. A Random Forest clas-
 238 sifier trained to identify the source sensor from the harmonized features alone
 239 reaches 99.8% accuracy against a 66.3% majority-class baseline (Fig. 2b); re-
 240 stricting it to the visible bands gives 97.9%, and to scale-invariant band ratios
 241 99.5%. This is a classifier two-sample test (Friedman, 2003; Lopez-Paz and
 242 Oquab, 2017): if two samples come from the same distribution, no classifier
 243 can separate them above chance. Sensor identity therefore remains fully en-
 244 coded in the harmonized reflectance, and any model trained on the pooled
 245 archive can read it. The test is not new, but to our knowledge it has not
 246 been used as a harmonization acceptance check; it costs one model fit and it
 247 is decisive.

248 Refitting locally on the 270 station–date events observed by both sensors
 249 gives water-specific coefficients (Table A.3). These restore the spectral sig-
 250 nature almost exactly — the median Landsat/Sentinel-2 ratio falls from 15.4
 251 to about 1.0 across all six bands (Fig. 2d) — but per-observation agreement
 252 remains uneven (Fig. 2c). Green recovers usable agreement ($r = 0.71$), and
 253 NIR and SWIR reach $r \approx 0.6$. Blue does not: $r = 0.26$. This is the most
 254 consequential result of the section, because SHAP attributes 72.6% of the
 255 retrieval to green/blue bands and ratios (Section 3.7). The band on which

256 the retrieval most depends is the band on which the two sensors least agree.

257 3.2. *The campaign, not the station, is the blocking unit*

258 Table 1 reports the retrieval under each validation design alongside the
259 null models (Fig. A.6). Blocking by station changes nothing relative to a ran-
260 dom split (both $R^2 = 0.596$), which under the usual reading would be taken
261 as evidence of no spatial leakage. The null models show why that reading
262 is wrong: under campaign-blocked folds the station-mean null still attains
263 $R^2 = 0.208$ and the zone-mean null $R^2 = 0.287$, whereas the campaign-date-
264 mean null collapses to -0.012 . Dependence in this dataset lives between
265 campaigns, not between stations, because a single campaign samples the
266 whole lake within a few days under one set of atmospheric and limnological
267 conditions. Station-blocked folds leave that structure intact.

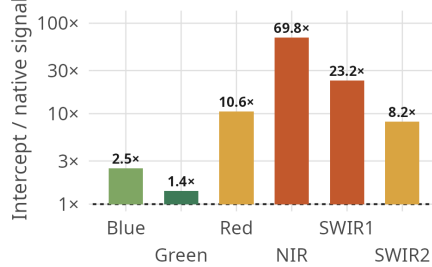
268 Under campaign-blocked validation — our primary design — the Random
269 Forest attains $R^2 = 0.574$ and RMSE 1.86 m over 812 match-ups, essentially
270 unbiased (-0.02 m). Blocking by year is stricter still ($R^2 = 0.495$). Collaps-
271 ing cross-sensor pseudo-replicates to 540 independent events leaves accuracy
272 unchanged ($R^2 = 0.571$).

273 3.3. *Retrieval accuracy and benchmarks*

274 Against this design the classical blue/green band ratio attains $R^2 = 0.097$
275 (RMSE 2.71 m) and a multiband linear model $R^2 = 0.323$ (RMSE 2.34 m)
276 (Table A.4, Fig. 3). We report one methodological caution alongside these
277 figures. An unbounded implementation of the same band-ratio algorithm —
278 log-transformed and back-transformed without constraining predictions to
279 the observable range — produces a single prediction of 143.5 m against an

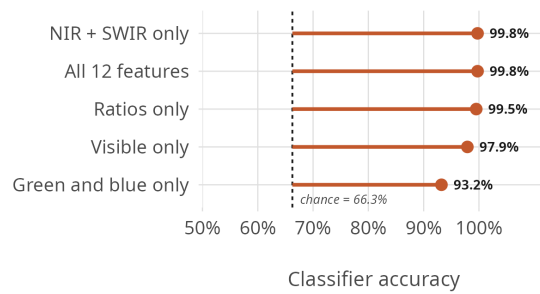
(a) Land intercept exceeds the signal

Roy et al. (2016) coefficients, fitted over vegetation and soil, vs native Landsat reflectance over the lake water. The dashed line marks intercept = signal



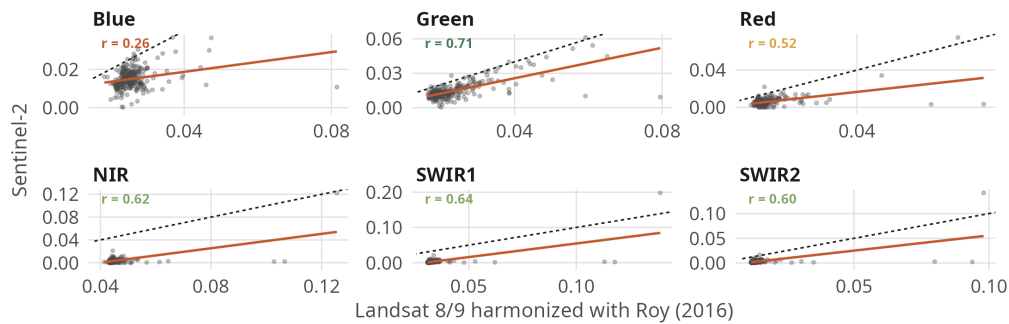
(b) Sensors remain distinguishable

Accuracy of a Random Forest predicting the source sensor from already harmonized reflectance. If harmonization worked, it would be at chance



(c) Blue is the band that agrees least between sensors

Sentinel-2 reflectance vs Roy-harmonized Landsat in the 270 events observed by both sensors at the same station and date. The dashed line is 1:1



(d) Local recalibration over water does harmonize bands

Median reflectance over water by sensor. Labels are the Landsat / Sentinel-2 ratio: it goes from 15.4x in NIR to ~1.0x in every band

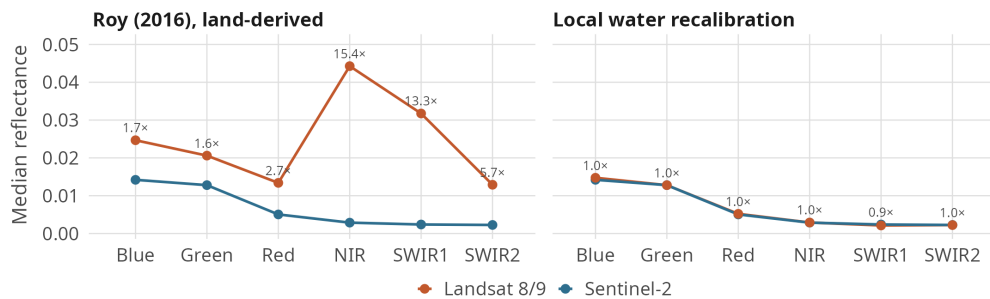


Figure 2: Why land-derived harmonization fails over clear water. (a) The Roy intercept relative to the native over-water Landsat signal, by band. (b) Accuracy of a classifier predicting the source sensor from harmonized features; under a successful harmonization it would sit at chance. (c) Sentinel-2 versus harmonized Landsat reflectance on the 270 paired station-date events; blue agrees worst. (d) Median spectral signature before and after local water recalibration; labels are the Landsat/Sentinel-2 ratio.

Table 1: Validation design, null models and blocking units (Random Forest, $n = 812$ unless noted). A blocking unit is effective only if the corresponding null model collapses under it.

	Setting	R^2	RMSE (m)
Null models	Zone-mean	0.287	2.40
	Station-mean	0.208	2.53
	Campaign-date-mean	-0.012	2.86
Blocking unit	Random K -fold	0.596	1.81
	By station	0.596	1.81
	By campaign date	0.574	1.86
	By year	0.495	2.02
	One record per event ($n = 540$)	0.571	1.77
Zone extrapolation	Leave out Lago Menor ($n = 133$)	0.247	1.90
	Leave out Bahía de Puno ($n = 88$)	0.036	2.11
	Leave out Lago Mayor ($n = 591$)	-0.047	2.52
Temporal	Train ≤ 2019 , test 2022 & 2024 ($n = 226$)	0.507	2.63

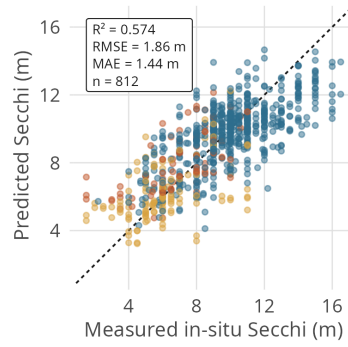
280 observed maximum of 16.5 m, which alone drives R^2 to -2.458 and RMSE to
281 5.29 m (Fig. 3d). Benchmarks of classical algorithms reported without phys-
282 ical bounds should be treated with caution: a single unconstrained extrapo-
283 lation manufactures an arbitrarily large advantage for the machine-learning
284 model.

285 3.4. *Model family and tuning do not change the picture*

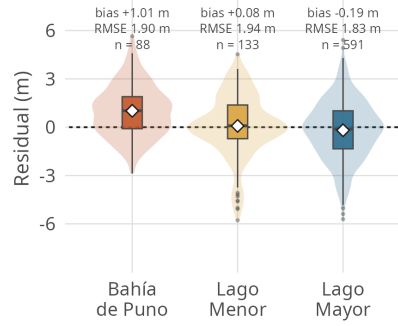
286 We compared four learner families, each tuned by Optuna inside nested
287 cross-validation, against target transformations and reduced feature sets (Ta-
288 ble A.5). The best configuration — an equal-weight ensemble of the four
289 tuned models — reached $R^2 = 0.583$, an improvement of 0.009 over the
290 untuned Random Forest. That gain does exceed the spread across random
291 seeds (0.003 in R^2), so it is not noise; but it corresponds to 0.02 m of RMSE
292 on a lake whose reference measurement is recorded to the nearest half metre,
293 and it changes no conclusion in this article. Neither log-transforming the
294 target nor tuning individual learners helped more than that, and restricting
295 the feature set cost accuracy.

296 We therefore retain the untuned Random Forest as the reported model:
297 it is simpler, faster and easier to reproduce, and reporting a tuned ensemble
298 for a 0.009 gain would overstate the precision of the product. The ceiling
299 here is set by the data — a dry-season-only record, a reference measurement
300 quantized at half a metre, and a target whose variance is dominated by a
301 three-level trophic gradient — not by the choice of learner.

(a) Predicted vs measured
Random Forest, campaign-blocked
validation. Dashed line is 1:1

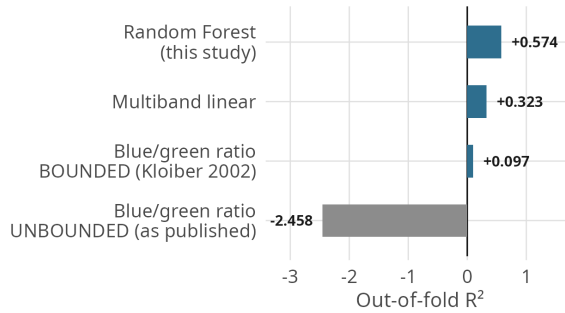


(b) Bias concentrates in the bay
Residuals (predicted - measured) by zone.
Diamond is the mean. Bahía de Puno, the
most eutrophic zone, +1.0 m



• Bahía de Puno • Lago Menor • Lago Mayor

(c) The baseline was misimplemented
Same validation design for all.
The published baseline undid the log without
bounds; a single 143 m prediction ruined its R^2



(d) One impossible value ruins R^2
Classical ratio on log scale.
The 143 m point is the one that produced
the $R^2 = -2.46$

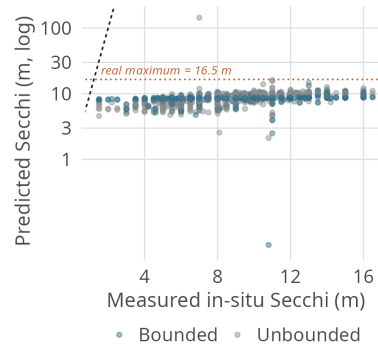


Figure 3: Retrieval under the primary design. (a) Out-of-fold predicted versus measured Secchi depth. (b) Residuals by trophic zone; the retrieval overestimates in eutrophic Bahía de Puno. (c) Benchmark against classical and linear baselines. (d) The unbounded classical implementation in log scale: one impossible prediction destroys the metric.

3.5. Where the retrieval stops working

Withholding an entire trophic zone collapses the retrieval (Table 1): R^2 falls to 0.247 for Lago Menor, 0.036 for Bahía de Puno and -0.047 for Lago Mayor. The model interpolates within optical regimes it has seen and does not extrapolate to a water type absent from training. On the temporal hold-out (train ≤ 2019 , test 2022 and 2024) it retains $R^2 = 0.507$, though 57 of the 69 test stations also appear in training (82.6%), so this measures temporal rather than fully independent generalization.

Error is unevenly distributed (Table A.6). The retrieval carries a positive bias of 1.01 m in Bahía de Puno, the most eutrophic and most vulnerable zone, because the clear-water majority pulls predictions upward.

Two features of the harmonized archive should be read together with these numbers. Removing the two bands most corrupted by the land-derived intercepts lowers R^2 only from 0.574 to 0.541; using scale-invariant ratios alone gives 0.468; and applying our water-specific recalibration gives 0.518. Training on each sensor separately gives 0.563 (Sentinel-2, $n = 274$) and 0.549 (Landsat, $n = 538$). The pooled model’s small advantage over these variants is consistent with it exploiting sensor identity as an additional covariate rather than with genuine cross-sensor synergy.

3.6. Calibrated per-prediction uncertainty

Split-conformal intervals are well calibrated *marginally* at every nominal level tested: empirical coverage is 50.9% at 50% nominal, 79.1% at 80%, 89.4% at 90% and 94.3% at 95%, a maximum calibration error of 0.94 percentage points (Fig. A.7a).

326 They are not calibrated *conditionally*, and this matters more for opera-
 327 tional use than the marginal result. Coverage falls to 71.3% in the clearest
 328 water ($Z_{SD} > 12.5$ m) and 79.5% in the most turbid ($Z_{SD} < 5$ m), while
 329 exceeding the nominal level in the middle of the range (Fig. A.7c). The
 330 mechanism is structural rather than incidental: split conformal derives a sin-
 331 gle global quantile of absolute residuals, so the interval has constant width
 332 (6.12 m at 90%) everywhere and cannot adapt to the local difficulty of the re-
 333 trieval. It therefore buys its marginal guarantee by over-covering in the well-
 334 sampled middle of the range and under-covering at both extremes — which
 335 are precisely where management decisions are made. A locally adaptive vari-
 336 ant such as conformalized quantile regression (Romano et al., 2019) would
 337 be required for reliable conditional coverage, and we recommend reporting
 338 coverage conditional on the predicted value as standard practice whenever
 339 conformal intervals accompany an operational product.

340 For a lake whose transparency spans 1.5–16.5 m, a ± 3 m interval is in-
 341 formative for zone-scale contrasts and multi-metre change, and is not infor-
 342 mative for sub-metre differences.

343 3.7. Interpretation and retrievability

344 Out-of-fold SHAP attributes 72.6% of the retrieval to green and blue
 345 bands and their ratios, 14.4% to NIR and SWIR, and the remainder to
 346 other indices (Fig. 4a). Green/blue alone accounts for 37%. The dependence
 347 is monotone and of the sign expected from water-clarity optics (Fig. 4b):
 348 a higher green-to-blue ratio pushes the prediction toward less transparent
 349 water. The retrieval is therefore grounded in optics rather than location
 350 memorization — while resting on the band pair with the worst cross-sensor

351 agreement.

352 With real labels and the same campaign-blocked design, no other vari-
353 able is retrievable (Fig. 4c): chlorophyll-*a* reaches $R^2 = 0.012$, suspended
354 solids 0.056, in-situ water temperature 0.113, and Landsat thermal surface
355 temperature 0.020 ($r = 0.14$, $n = 662$). Their in-situ ranges — chlorophyll-*a*
356 0–4.6 mg m⁻³, suspended solids mostly below 10 mg L⁻¹ — lie below the op-
357 tical and thermal detection limits of these sensors. We subject the negative
358 claim to the same design as the positive one, and report it to caution against
359 applying chlorophyll algorithms (Gitelson et al., 2009; Mishra and Mishra,
360 2012) or thermal retrievals (Ruiz-Verdú et al., 2016) in clear high-altitude
361 water without local validation.

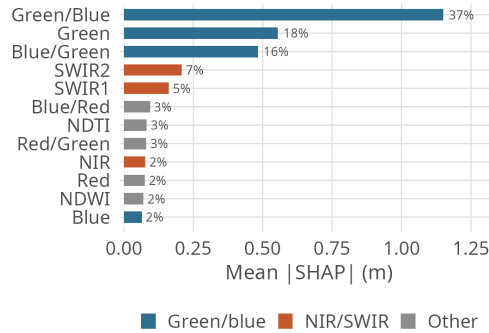
362 3.8. Temporal patterns and the fragility of trend claims

363 Mann–Kendall tests on annual median in-situ transparency over the full
364 2011–2024 record show no statistically significant trend in any zone (Bahía
365 de Puno: Sen slope 0.16 m yr⁻¹, $p = 0.241$; Lago Mayor: 0.25 m yr⁻¹,
366 $p = 0.138$; Fig. A.8a).

367 That conclusion is fragile in a way worth reporting explicitly. Restricting
368 the series to 2013 onwards — the natural choice if trends were computed
369 on the satellite match-up subset, whose record begins with Landsat 8 —
370 turns both Bahía de Puno ($p = 0.017$) and Lago Mayor (Sen +0.42 m yr⁻¹,
371 $p = 0.012$) significantly positive. Across start years from 2011 to 2014, 5 of
372 12 zone–start-year combinations change verdict (Fig. A.8b). The year 2013
373 has the lowest annual medians on record, so beginning there manufactures
374 an upward trend. With 9–11 annual values and three gap years the test has
375 low power, and the correct reading is absence of evidence, not evidence of

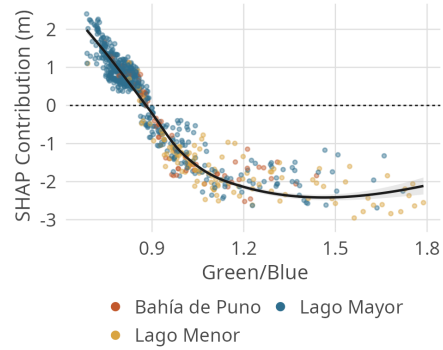
(a) Green and blue dominate attribution

Mean |SHAP| computed OUT-OF-FOLD: each observation is explained by the model that did not see it. Labels are the share of the total



(b) How Green/Blue acts

SHAP value vs. feature value. Greener water relative to blue pushes the prediction toward less transparent water. Trimmed to the 1-99th percentile (18 extreme points out)



(c) Only transparency is retrievable in this water

Random Forest with the same honest design (campaign blocking) for each variable. The negative claim is subject to the same rigor as the positive one

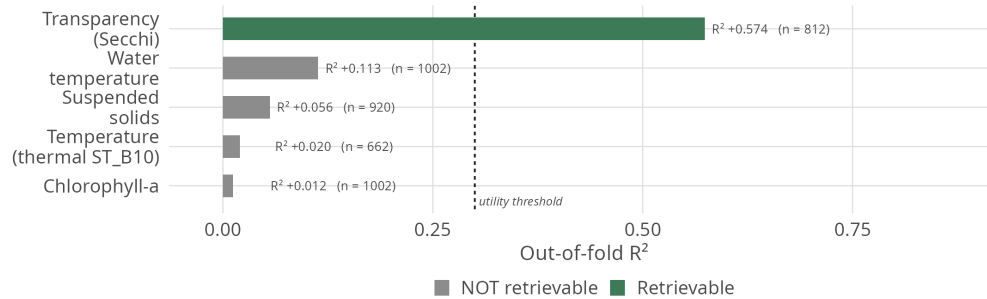


Figure 4: Interpretation and operational envelope. (a) Out-of-fold SHAP importance; green and blue dominate. (b) SHAP dependence on the leading feature, by trophic zone. (c) Retrievability of each water-quality variable under the same campaign-blocked design; only transparency clears a useful threshold.

376 stability. We recommend that trend analyses in campaign-based lake records
377 report this sensitivity as a matter of course.

378 4. Discussion

379 4.1. Implications for harmonized multi-mission archives

380 The virtual-constellation approach (Claverie et al., 2018; Pahlevan et al.,
381 2019) has become the default for multi-decadal lake monitoring, and the Roy
382 et al. (2016) coefficients are its most reused transform. Our results place
383 a boundary on where that default is safe. Those coefficients were fitted on
384 terrestrial targets and carry additive intercepts appropriate to surfaces one
385 to two orders of magnitude brighter than clear water. Over Lake Titicaca the
386 intercept exceeds the native over-water signal by a factor of 69.8 in the near-
387 infrared and by 2.5 even in blue, and the two sensors remain 99.8% separable
388 after the transform. That water reflectance from these products is unreliable
389 is already documented (Pahlevan et al., 2019; Masek et al., 2021); what our
390 paired observations add is the magnitude — the problem is not confined
391 to the near-infrared, and it is large enough to make the harmonized bands
392 function as sensor labels for a machine-learning model — and the finding that
393 local recalibration repairs the aggregate spectral signature without repairing
394 per-observation agreement in blue.

395 The practical consequence is direct. A machine-learning model trained
396 on a pooled, land-harmonized archive can achieve apparently good accuracy
397 while having learned two sensor-specific mappings joined by a readable sensor
398 label. That is not a harmonized retrieval, and it will not behave as one when
399 applied to a scene from either sensor alone. The diagnostic is cheap and we

400 recommend it as routine: train a classifier to predict the source sensor from
401 the harmonized features and report its accuracy against the majority-class
402 baseline. If the sensors remain separable, the harmonization has not done
403 its job. For clear oligotrophic lakes we further recommend deriving water-
404 specific coefficients from same-day paired observations, reporting per-band
405 agreement, and — where blue-band agreement is as poor as we find here —
406 reporting single-sensor retrievals alongside any pooled product.

407 *4.2. Implications for validation design*

408 Blocking by station is the standard remedy for spatial autocorrelation
409 in this literature, and in our data it changes nothing: random and station-
410 blocked folds give identical accuracy. Read conventionally, that would be
411 reassuring. The null models show it is not. Under campaign-blocked folds
412 the station-mean null still explains 21% of variance and the zone-mean null
413 29%, while the campaign-date-mean null explains none. In a campaign-based
414 monitoring programme the whole lake is sampled within a few days under
415 shared atmospheric and limnological conditions, so observations cluster by
416 campaign far more strongly than by location; station-blocked folds leave that
417 structure entirely intact.

418 The general point is that the blocking unit should be diagnosed, not
419 assumed. Reporting null models under each candidate design costs little
420 and identifies the unit that carries the dependence. This applies well beyond
421 Lake Titicaca: most large-lake monitoring programmes sample in campaigns,
422 so the same mismatch between the conventional blocking unit and the real
423 dependence structure is likely to be widespread.

424 For our retrieval, the honest figure is $R^2 = 0.574$ (RMSE 1.86 m) under

425 campaign-blocked validation, against 0.287 for a zone-mean null: the model
426 roughly doubles the skill available from knowing only which basin a sample
427 came from. That is a real but modest contribution, and stating it that way
428 is more useful to a manager than an unqualified R^2 .

429 4.3. Operational envelope for Lake Titicaca

430 Our RMSE of 1.86 m spans roughly 12% of the observed Z_{SD} range
431 and sits below the published range for Z_{SD} retrieval ($R^2 \approx 0.6$ – 0.9 ; Zhang
432 et al., 2022; Lee et al., 2015), obtained here under a predominantly clear
433 oligotrophic regime and a stricter validation design. It is also approaching
434 the granularity of the reference measurement itself: disk readings are recorded
435 to the nearest half metre, giving only 73 distinct values across 812 match-ups.

436 Four constraints bound operational use. The retrieval carries a 1.01 m
437 positive bias in Bahía de Puno — the most eutrophic and most vulnerable
438 zone — so mapped values there must be read with their intervals and con-
439 firmed by field sampling. Leave-one-zone-out results show the model cannot
440 extrapolate to an unsampled optical regime; integration must be incremen-
441 tal, each new campaign both validating the current map and extending the
442 training set. The retrieval is calibrated and validated for the austral dry sea-
443 son only, and the reason is worth stating precisely because it determines the
444 remedy. It is not that wet-season imagery is unusable at 3810 m, nor that our
445 match-up criteria excluded it: it is that the field programme has produced
446 a single wet-season campaign in fourteen years, and that campaign predates
447 both satellites. Wet-season blooms in Bahía de Puno — the phenomenon a
448 transparency product would most usefully warn about — are therefore en-
449 tirely unrepresented, and the model should not be applied between December

450 and March.

451 This converts the limitation into a specific, actionable recommendation
452 for the monitoring authorities. Extending the protocol to the wet season does
453 not require a better algorithm; it requires field data that do not yet exist. A
454 single additional wet-season campaign per year, timed to any cloud-free win-
455 dow and covering the three trophic zones, would within a few years supply
456 enough independent campaign-blocks to calibrate and validate a wet-season
457 retrieval under the same design used here. Because the binding constraint is
458 sampling rather than sensing, this is among the cheapest available improve-
459 ments to Lake Titicaca’s water-quality monitoring capability. And because
460 conformal coverage degrades at both extremes of the transparency range, the
461 nominal 90% interval should be treated as optimistic in the clearest and most
462 turbid water.

463 Within those bounds the product is useful to the regional monitoring
464 authorities (IMARPE, ALT, OEFA, and the binational management frame-
465 work). A calibrated ± 3 m interval resolves zone-scale contrasts and multi-
466 metre change, which is the scale at which clarity loss in Bahía de Puno
467 would be actionable, and a single transboundary product under one cali-
468 bration supports comparable assessment across the Peru–Bolivia boundary
469 where coordinated field monitoring is logistically difficult. It does not resolve
470 sub-metre change and should not be presented as doing so.

471 4.4. *Limitations*

472 Beyond the constraints already stated, three limitations bound this work.
473 The in-situ record is episodic, with no campaigns in 2020, 2021, 2023 and only
474 one wet-season campaign in fourteen years, so the temporal hold-out rests

475 on two test years, the trend analysis on 9–11 annual values, and the seasonal
476 envelope on none. Transparency above approximately 15 m saturates the
477 optical signal. And this is a single lake: the two diagnostics we propose are
478 general, but the water-specific coefficients in Table A.3 are local to Titicaca
479 and must be refitted elsewhere.

480 5. Conclusions

481 Using 1002 satellite–field match-ups over twelve years, we show that two
482 practices imported from the turbid-water remote-sensing literature fail in a
483 clear high-altitude large lake. Cross-sensor harmonization coefficients derived
484 over land leave Sentinel-2 and Landsat 99.8% separable over oligotrophic
485 water, because their additive intercepts exceed the native over-water signal
486 by up to a factor of 69.8; and validation blocked by monitoring station leaves
487 the dominant dependence — between campaigns — entirely intact.

488 Correcting both, a Random Forest retrieves Secchi depth with $R^2 = 0.574$
489 and RMSE 1.86 m, roughly double the skill of a zone-mean null, with confor-
490 mal prediction intervals that are calibrated marginally but under-cover at the
491 extremes of the transparency range. Skill collapses under extrapolation to an
492 unsampled trophic zone; chlorophyll-*a*, suspended solids and thermal tem-
493 perature are not retrievable at all; and no transparency trend is significant
494 once the full in-situ record is used rather than the satellite-era subset.

495 We therefore recommend two routine checks for multi-mission lake water-
496 quality products, neither of which requires new methodology: report the
497 accuracy of a classifier two-sample test on the harmonized features, and re-
498 port null models under each candidate validation blocking unit. Both cost

one model fit, and both distinguish genuine optical retrieval from sensor labelling and sampling design. For Lake Titicaca the resulting product is a dry-season, zone-scale transparency indicator with honest ± 3 m intervals — less than has sometimes been claimed for such lakes, and more than the region has had.

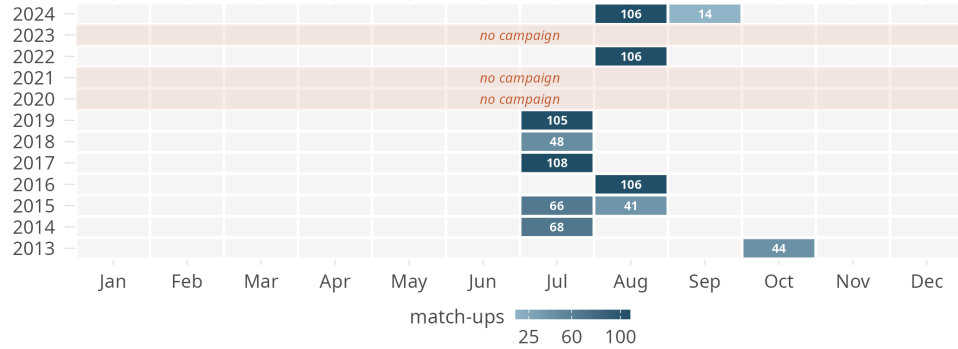
Appendix A. Supplementary Figures and Tables

Table A.2: Effect of the Roy et al. (2016) OLI→MSI transform over Lake Titicaca water. Native OLI is the Collection-2 surface reflectance recovered by inverting the transform. Reflectance is dimensionless.

Band	Roy intercept a	Native OLI median	Intercept / signal	Harmonized LS median	Harmonized / S2 median
Blue	0.0183	+0.0072	2.5	0.0247	1.74
Green	0.0123	+0.0089	1.4	0.0206	1.61
Red	0.0123	+0.0013	10.6	0.0134	2.66
NIR	0.0448	−0.0006	69.8	0.0443	15.42
SWIR1	0.0306	+0.0013	23.2	0.0317	13.34
SWIR2	0.0116	+0.0013	8.2	0.0129	5.70

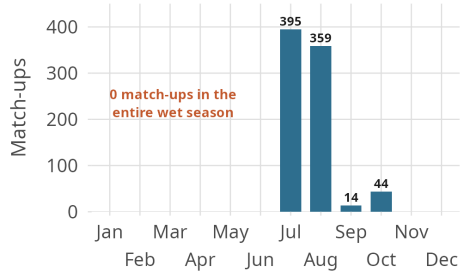
(a) Three years with no campaigns and no data outside the dry season

Number of satellite-field match-ups by year and month. There is no data for 2020, 2021 or 2023, so the 2022-2024 hold-out only contains 2022 and 2024



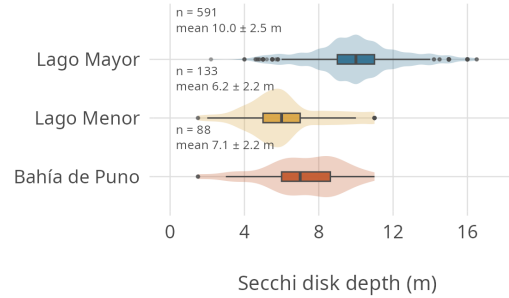
(b) Only dry season data exists

Bahía de Puno blooms occur during the wet season, which the dataset does not cover at all



(c) The trophic gradient dominates Secchi

Only 73 distinct values in 812 match-ups: readings are rounded to half a metre, so the RMSE of 1.86 m borders the granularity of the reference measurement itself



■ Wet season (Dec-Mar) ■ Transition ■ Dry season (Jul-Oct)

Figure A.5: What the dataset actually contains. (a) Match-ups by year and month; three years have no campaign at all, so a hold-out described as “2022–2024” contains only 2022 and 2024. (b) Seasonal coverage: zero match-ups in the wet season, when blooms occur in Bahía de Puno. (c) Secchi distribution by trophic zone; the reference measurement is quantized at roughly half a metre.

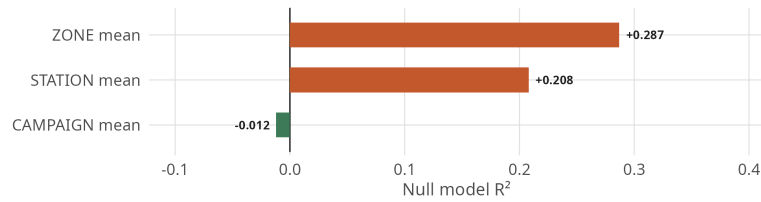
Table A.3: Water-specific LS→S2 recalibration fitted on 270 paired station–date observations, of the form $\rho_{S2} = a + b \rho_{LS, Roy}$.

Band	a	b	Pearson r	Agreement
Blue	+0.0084	0.258	0.26	unusable
Green	−0.0009	0.665	0.71	good
Red	−0.0005	0.427	0.52	weak
NIR	−0.0250	0.631	0.62	acceptable
SWIR1	−0.0223	0.770	0.64	acceptable
SWIR2	−0.0057	0.615	0.60	acceptable

(a) Only campaign blocking neutralizes its own null model

R^2 of models predicting only the group mean, with folds blocked by campaign.

A blocking unit is effective if and only if its own null model collapses to zero under it



(b) Station blocking changes nothing; extrapolation breaks everything

Out-of-fold R^2 of the Random Forest; the large point is the primary design.

The fact that station blocking matches with random does not prove absence of leakage: it proves that it retains nothing

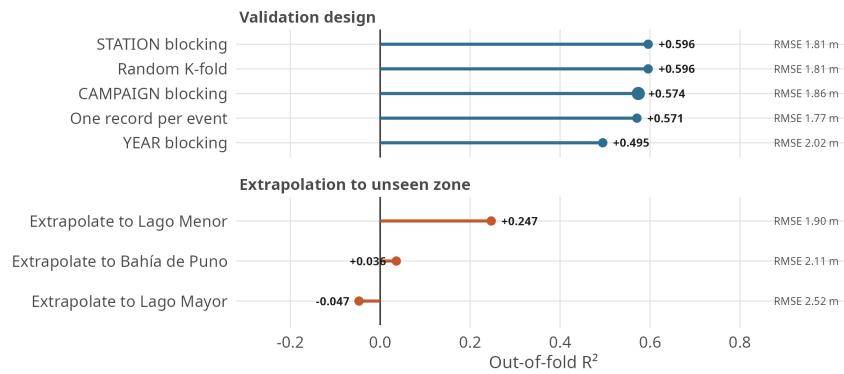
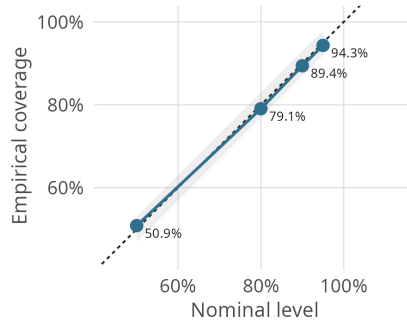
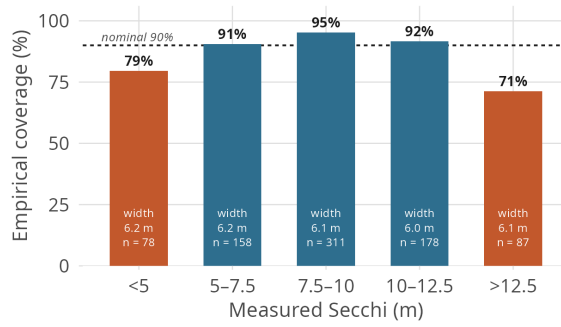


Figure A.6: Diagnosing the blocking unit. (a) Null models evaluated under campaign-blocked folds: only the campaign-mean null collapses, so only campaign blocking is doing work. (b) Out-of-fold R^2 under each validation design and under extrapolation to an unseen trophic zone. That station blocking matches the random split is not evidence of no leakage; it is evidence that station blocking retains nothing.

(a) Calibrated, on average
Out-of-fold empirical coverage vs nominal. Gray band is ± 3 points



(c) But not at the ends of the range
Split-conformal uses a single global quantile, so the width is constant (~ 6.1 m) and the interval cannot adapt: it undercovers in the clearest water



(b) The 90% interval vs the measurement
Match-ups ordered by measured transparency. The line is the prediction, the band is the 90% conformal interval. 86 out of 812 points fall outside

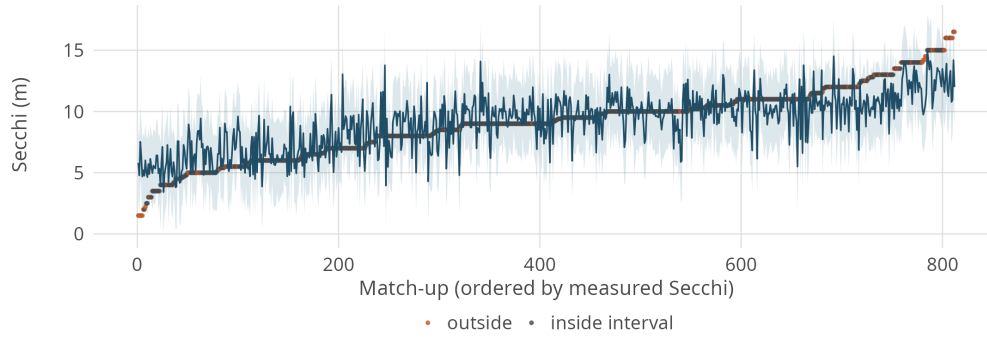
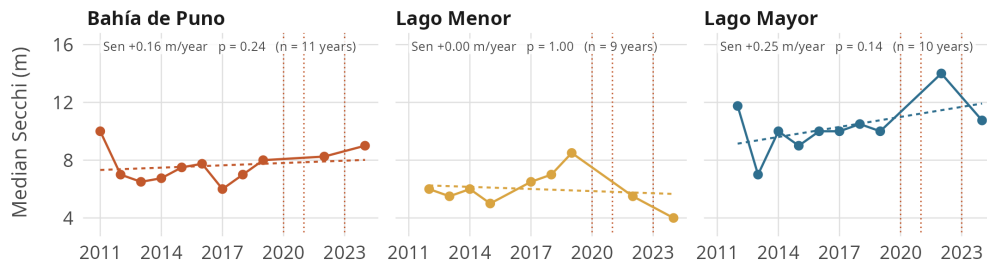


Figure A.7: Conformal prediction intervals under campaign-blocked validation. (a) Empirical versus nominal coverage: marginally calibrated across the 50–95% range. (b) The 90% interval against the measured value. (c) Coverage conditional on measured transparency. Because split conformal uses a single global residual quantile, the interval width is constant (≈ 6.12 m) and cannot adapt, so the marginal guarantee is bought by over-covering in the middle of the range and under-covering at both extremes.

(a) No zone shows significant trend in the full record

Annual median transparency in-situ (2011–2024, 734 readings). The vertical dotted lines mark the years with no campaign: 2020, 2021 and 2023



(b) The verdict is set by the start year, not the lake

Same Mann-Kendall test, same data, different start year. Starting in 2013 — the year with the lowest medians in the record — turns two zones into "significantly more transparent"

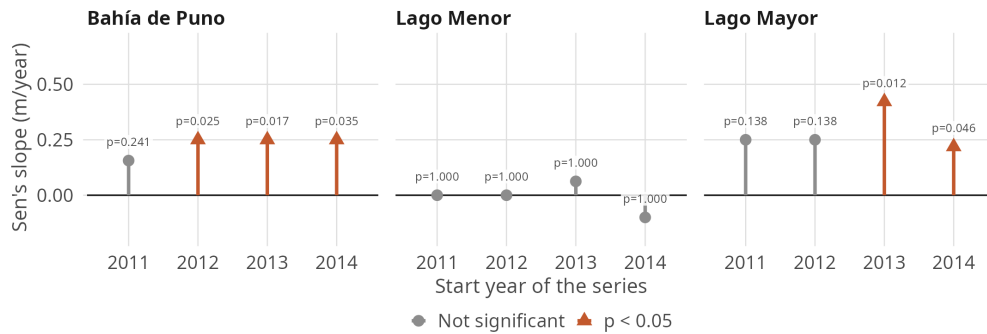


Figure A.8: Transparency trends and their fragility. (a) Annual median in-situ transparency by zone over the full 2011–2024 record, with Mann–Kendall/Sen statistics; no trend is significant. (b) The same test with different start years: beginning the series in 2013, the lowest-median year on record, turns two zones significantly positive.

Table A.4: Head-to-head benchmark, campaign-blocked GroupKFold, $n = 812$.

Algorithm	R^2	RMSE	MAE	Bias
Classical blue/green, unbounded ^a	−2.458	5.29	1.91	−0.19
Classical blue/green, bounded ^b	0.097	2.71	2.13	−0.46
Multiband linear	0.323	2.34	1.78	−0.01
Random Forest (this study)	0.574	1.86	1.44	−0.02

RMSE, MAE and bias in metres.

^a back-transformed without bounding predictions; see text.

^b standard form of Kloiber et al. (2002).

Table A.5: Model selection under nested cross-validation (Optuna inside an inner campaign-blocked split of each outer training partition). Selected configurations; the full grid is in the repository.

Model	Features	R^2	RMSE (m)
Ensemble of the four tuned models	all 12	0.583	1.84
LightGBM, tuned	all 12	0.581	1.84
XGBoost, tuned	all 12	0.577	1.85
Random Forest, tuned	all 12	0.576	1.85
Random Forest, untuned (reported)	all 12	0.574	1.86
Extremely Randomized Trees, untuned	all 12	0.573	1.86
XGBoost, tuned, $\log Z_{SD}$	all 12	0.561	1.89
LightGBM, tuned	visible only	0.544	1.92
LightGBM, tuned	ratios only	0.488	2.04
Seed-to-seed spread of the untuned Random Forest: R^2 range 0.003 over five seeds.			

Table A.6: Out-of-fold error by trophic zone, campaign-blocked validation.

Zone	R^2	RMSE (m)	Bias (m)	n
Bahía de Puno	0.221	1.90	+1.01	88
Lago Menor	0.211	1.94	+0.08	133
Lago Mayor	0.448	1.83	−0.19	591

505 **CRedit authorship contribution statement**

506 **Dina Maribel Yana-Yucra:** Conceptualization, Data curation, For-
507 mal analysis, Investigation, Methodology, Project administration, Resources,
508 Software, Supervision, Validation, Visualization, Writing – original draft,
509 Writing – review & editing. **Richar Andre Vilca-Solorzano:** Conceptual-
510 ization, Data curation, Formal analysis, Investigation, Methodology, Project
511 administration, Resources, Software, Validation, Visualization, Writing –
512 original draft, Writing – review & editing. **Fred Torres-Cruz:** Conceptual-
513 ization, Data curation, Formal analysis, Investigation, Methodology, Project
514 administration, Resources, Software, Supervision, Validation, Visualization,
515 Writing – original draft, Writing – review & editing. **Vladimiro Ibañez:**
516 Conceptualization, Data curation, Formal analysis, Investigation, Methodol-
517 ogy, Project administration, Resources, Supervision, Validation, Visualiza-
518 tion, Writing – original draft, Writing – review & editing.
519 All four authors contributed substantively across every stage of the study,
520 and all read and approved the final manuscript.

521 **Declaration of competing interest**

522 The authors declare that they have no known competing financial inter-
523 ests or personal relationships that could have appeared to influence the work
524 reported in this paper.

525 **Data availability**

The satellite–field match-up datasets and the complete analysis pipeline, including the scripts that generate every number, table and figure in this article, are openly available at:

526 <https://github.com/Andre031222/harmonized-titicaca-transparency>



Repository

Raw imagery is freely available via Google Earth Engine; the in-situ measurements derive from published IMARPE/ALT and OEFA monitoring records.

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